

The Average-Reward Policy Gradient Theorem

The Bellman Route

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1 Average-Reward Policy Gradient Theorem

The objective is the long-run reward rate,

$$J_{avR}(\theta) = r(\pi_\theta) = \lim_{h \rightarrow \infty} \frac{1}{h} \sum_{t=1}^h \mathbb{E}^{\pi_\theta}[R_t | S_0] = \sum_s \mu_{\pi_\theta}(s) \sum_a \pi_\theta(a | s) r(s, a),$$

the time-average of the reward, which under ergodicity equals the one-step reward averaged over the stationary state-action distribution. The chain induced by π_θ is ergodic with stationary distribution μ_{π_θ} , and the differential action-value is q_{π_θ} . We assume finite $\mathcal{S}, \mathcal{A}$, $\pi_\theta(a | s)$ differentiable in θ , ergodicity, and bounded reward, so that μ_{π_θ} and q_{π_θ} are differentiable in θ . The target is

$$\nabla_\theta J_{avR}(\theta) = \nabla_\theta r(\pi_\theta) = \sum_s \mu_{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a | s) q_{\pi_\theta}(s, a).$$

Step 1: rearranged differential Bellman identity. The differential Bellman equation with the rate isolated on the left, valid for every (s, a) :

$$r(\pi_\theta) = r(s, a) - q_{\pi_\theta}(s, a) + \sum_{s'} p(s' | s, a) \sum_{a'} \pi_\theta(a' | s') q_{\pi_\theta}(s', a').$$

Step 2: differentiate. The reward $r(s, a)$ and transition p carry no θ , so $\nabla_\theta r(s, a) = 0$ and ∇_θ passes through p . The product rule on $\sum_{a'} \pi_\theta q_{\pi_\theta}$ gives

$$\nabla_\theta r(\pi_\theta) = -\nabla_\theta q_{\pi_\theta}(s, a) + \sum_{s'} p(s' | s, a) [A(s') + B(s')],$$

$$A(s') = \sum_{a'} \nabla_\theta \pi_\theta(a' | s') q_{\pi_\theta}(s', a'), \quad B(s') = \sum_{a'} \pi_\theta(a' | s') \nabla_\theta q_{\pi_\theta}(s', a').$$

A is the reinforce term; B is the ripple term.

Step 3: average against $\mu_{\pi_\theta}(s) \pi_\theta(a | s)$. The left side is constant in (s, a) and the weights sum to one, so it stays $\nabla_\theta r(\pi_\theta)$. The right side splits into three:

$$\begin{aligned} \nabla_\theta r(\pi_\theta) &= T_1 + T_2 + T_3, \\ T_1 &= -\sum_s \mu_{\pi_\theta}(s) \sum_a \pi_\theta(a | s) \nabla_\theta q_{\pi_\theta}(s, a), \\ T_2 &= \sum_s \mu_{\pi_\theta}(s) \sum_a \pi_\theta(a | s) \sum_{s'} p(s' | s, a) A(s'), \end{aligned}$$

$$T_3 = \sum_s \mu_{\pi_\theta}(s) \sum_a \pi_\theta(a | s) \sum_{s'} p(s' | s, a) B(s').$$

Step 4: the cancellation. Sum the (s, a) part of T_3 first, using the stationarity identity

$$\sum_s \mu_{\pi_\theta}(s) \sum_a \pi_\theta(a | s) p(s' | s, a) = \mu_{\pi_\theta}(s').$$

Then

$$T_3 = \sum_{s'} \mu_{\pi_\theta}(s') B(s') = \sum_{s'} \mu_{\pi_\theta}(s') \sum_{a'} \pi_\theta(a' | s') \nabla_\theta q_{\pi_\theta}(s', a') = -T_1,$$

after renaming $s' \rightarrow s, a' \rightarrow a$. So $T_1 + T_3 = 0$: the $\nabla_\theta q_{\pi_\theta}$ terms cancel, because μ_{π_θ} is invariant.

Step 5: what survives. Only T_2 remains. Collapse its (s, a) sums by the same identity:

$$\nabla_\theta r(\pi_\theta) = \sum_{s'} \mu_{\pi_\theta}(s') A(s') = \sum_{s'} \mu_{\pi_\theta}(s') \sum_{a'} \nabla_\theta \pi_\theta(a' | s') q_{\pi_\theta}(s', a').$$

Renaming $s' \rightarrow s, a' \rightarrow a$ gives the theorem.

$$\nabla_\theta J_{avR}(\theta) = \sum_s \mu_{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a | s) q_{\pi_\theta}(s, a) = \mathbb{E}_{S \sim \mu_{\pi_\theta}, A \sim \pi_\theta(\cdot | S)} [\nabla_\theta \log \pi_\theta(A | S) q_{\pi_\theta}(S, A)],$$

the second form by $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$. ■

Two things to notice. There is no $\frac{1}{1-\gamma}$ factor: the rate is already per-step, and μ_{π_θ} has total mass 1 (the discounted visitation η had mass $\frac{1}{1-\gamma}$). And $\nabla_\theta \mu_{\pi_\theta}$ never appears: differentiating $r(\pi_\theta) = \sum_s \mu_{\pi_\theta}(s) \sum_a \pi_\theta(a | s) r$ directly would have forced it, but starting from the Bellman identity dodged it. The visitation gradient vanishes silently in the Step 4 cancellation, exactly as $\nabla_\theta \eta$ vanished in the discounted recursive route. The one assumed step is that $\nabla_\theta q_{\pi_\theta}$ is well-defined and finite, which the regularity assumptions provide.